Images of the Vitali transversal under distinct rational rotations are disjoint

The Open Logic Project contributors

openlogic:set-theory/choice:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own lemma environment vitalinooverlap (source lines 1022-1024, the \ollabel on line 1023); the proof environment begins two lines below it and is its own proof, closing at source line 1034.

Editorial note. Neither the statement nor the proof contains an \olref or a \cite, so no source pointer and no bibliography entry are needed; the notation for the rotation group, composition and images used in the proof is declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained lemma prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's axiom-of-choice unit, counted once: distinct rational rotations of the Vitali transversal have disjoint images.

Allowed standard prerequisites (editorial).

- The transversal C and the equivalence relation on the reals. The relation on real numbers that makes r equivalent to s when r - s is rational is an equivalence relation, and C is a set selecting exactly one representative from each equivalence class.. The defining property of C is used at source lines 1031-1032 to pass from r_1 equivalent to r_2 to r_1 = r_2.
- The group of rational rotations. The set of maps rho_r : x maps to x + r for rational r, with composition and inversion, is a group that preserves the equivalence relation.. The proof forms the composite of rho_1 with the inverse of rho_2 and observes that it lies in the group (source lines 1029-1031), which is what turns the two representations of s into an equivalence between r_1 and r_2.

Lemma 0.1. *If* $\rho_1 \neq \rho_2$ *then* $\mathrm{im}(\rho_1)C \cap \mathrm{im}(\rho_2)C = \emptyset$.

Proof. Suppose $s \in \mathrm{im}(\rho_1)C \cap \mathrm{im}(\rho_2)C$. So $s = \rho_1(r_1) = \rho_2(r_2)$ for some $r_1, r_2 \in C$. Hence $\rho_2^{-1}(\rho_1(r_1)) = r_2$, and $\circ\rho_1\rho_2^{-1} \in \mathrm{SO}(2)$, so $r_1 \sim r_2$. So $r_1 = r_2$, as C selects exactly one member from each equivalence class under $\sim$. So $s = \rho_1(r_1) = \rho_2(r_1)$, and hence $\rho_1 = \rho_2$. □